through time. Right now, the watch on your wrist and the clock on the wall are ticking away, showing that you and everything around you are relentlessly moving through time, relentlessly moving from one second to the next and the next. Newton thought that motion through time was totally separate from motion through space—he thought these two kinds of motion had nothing to do with each other. But Einstein found that they are intimately linked. In fact, *the* revolutionary discovery of special relativity is this: When you look at something like a parked car, which from your viewpoint is stationary—not moving through space, that is—*all* of its motion is through time. The car, its driver, the street, you, your clothes are all moving through time in perfect synch: second followed by second, ticking away uniformly. But if the car speeds away, some of its motion through time is *diverted* into motion through space. And just as Bart's speed in the northward direction slowed down when he diverted some of his northward motion into eastward motion, the speed of the car through *time* slows down when it diverts some of its motion through time into motion through *space.* This means that the car's progress through time slows down and therefore *time elapses more slowly for the moving car and its driver than it elapses for you and everything else that remains stationary.*

That, in a nutshell, is special relativity. In fact, we can be a bit more precise and take the description one step further. Because of the retrofitting, Bart had no choice but to limit his top speed to 65 miles per hour. This is important to the story, because if he sped up enough when he angled northeast, he could have compensated for the speed diversion and thereby maintained the same net speed toward the north. But with the retrofitting, no matter how hard he revved the skateboard's engine, his total speed—the combination of his speed toward the north and his speed toward the east—remained fixed at the maximum of 65 miles per hour. And so when he shifted his direction a bit toward the east, he necessarily caused a decreased northward speed.

Special relativity declares a similar law for all motion: *the combined speed of any object's motion through space and its motion through time is always precisely equal to the speed of light.* At first, you may instinctively recoil from this statement since we are all used to the idea that nothing but light can travel at light speed. *But that familiar idea refers solely to motion through space.* We are now talking about